CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 9: Matrices

Every formula and result of the chapter on one sheet

1	Order of a matrix	Rows $\times$ columns: $m \times n$
2	Equal matrices	Same order and equal corresponding elements
3	Addition and subtraction	Same order only; add or subtract corresponding elements
4	Scalar multiplication	$k\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$
5	Condition for AB to exist	Columns of A = rows of B
6	Order of the product	$(m \times n)(n \times p) = m \times p$
7	Product of 2×2 matrices	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} ap+br & aq+bs \\ cp+dr & cq+ds \end{bmatrix}$
8	Identity matrix	$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; AI = IA = A$
9	Zero matrix	$A + O = A; AO = OA = O$
10	Not commutative	$AB \neq BA$ in general
11	Distributive law	$A(B + C) = AB + AC$
12	Transpose	Rows and columns interchanged: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^t = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$